DeployHub AI assistant

What it does, how it is built, what it costs, the risks with their fixes, and the bugs found while building · 28 October 2026

Status: built and tested on 28 October 2026. Every decision here was agreed in chat. Where the build differs from the first plan, this version says so (marked **Changed**). It runs now with simple built-in answers; real AI answers start when you add a free GEMINI_API_KEY to deployhub-backend/.env. **Changed (same day):** Google Gemini (free tier) is now the default AI; Claude stays available with one setting. Bugs found while building are in section 11 and in bugreport.pdf (BUG-32 to BUG-34).

1. What the assistant is

A chat helper inside DeployHub, like Razorpay's "RAY" on its sign-up page. It talks to the user in plain language, collects their details step by step, checks them, and suggests a fix when something is wrong. After sign-up it stays available as a deploy helper, and it opens by itself when something fails.

- **Sign-up assistant:** chat on the left, a "Your details" card on the right that fills in with green ticks.
- **Student check:** students can verify with a student ID photo (or a college email code) and get **Rs 5 off** each paid deploy.
- **Deploy helper:** help with the New project form and answers to questions.
- **Pops up on problems:** explains failed deploys, missing settings, crashes, failed payments and refused database imports, and says what to change.

2. Decisions made

Topic	Decision	Why
Where the code lives	Inside deployhub-backend, new **assistant** package (Java 21)	Uses DeployHub's login, checks and logs directly; one app to run and deploy at M7
Language	Java (backend) + TypeScript/Angular (chat panel)	Python was considered; it would need a separate service, a newer Python and a third app to run
AI provider	**Google Gemini** (default, free tier) or Claude from Anthropic: one setting	You asked for a free option; Claude stays ready for when quality matters more than cost
Models	Gemini: **3.5 Flash-Lite** for chat, **3.8 Flash** for failed deploys. Claude: Haiku 5.5 / Opus 5.5	A fast model for chat, a smarter one for long, messy deploy logs
Details collected	Name, email (confirmed with a code), student or working professional, password	Enough for an account; nothing DeployHub doesn't need
Aadhaar	**Not collected**	See section 7: legal limits and leak risk; DeployHub has no need for it
Students	Optional student ID photo or college email code; Rs 5 off each paid deploy (Rs 49 -> Rs 44)	The 2 free projects stay free
API key	GEMINI_API_KEY (or ANTHROPIC_API_KEY) in deployhub-backend/.env (you add it)	A Claude subscription (Pro/Max) can't be used for an app's users; a Gemini key is free
Free tier and ID photos	On Gemini's free tier, student ID photos are **not sent**; students use their college email	Google says free-tier content is used to improve its products

3. Sign-up conversation, step by step

Step	Assistant asks	Checks and suggestions
1	"Hi! What's your name?"	Letters and spaces only (any language, e.g. Hindi), no numbers or special characters; at most 100 characters; extra spaces removed
2	"What's your email?"	Format; common typos (gmial.com -> "Did you mean gmail.com?" with buttons "Yes, use ..." / "No, keep mine"; the user decides)
3	"I've sent a 6-digit code to your email."	Same design as Forgot password (10 min, 5 tries). Already registered -> Log in / Forgot password links. "Send a new code", "Use another email"

Step	Assistant asks	Checks and suggestions
4	"Are you a student or a working professional?"	Two buttons, not free text
5	"Choose a password" (two normal hidden boxes)	Same password rules as today; **never sent to the AI** and never shown in the chat
6	Students only, after the account exists: "Upload a photo of your student ID"	Optional: "Later" -> Settings > Student discount (where the college email way is too)
7	"All set! Shall we deploy your first project?"	Buttons: Deploy my first project / Go to dashboard

The right-hand card shows each detail with a green tick when it passes. A message ending in "?" is a question for the assistant ("Why do you need my email?"); it answers and then asks the step again. Name and email are checked by DeployHub's own code, so sign-up costs nothing and works without the AI. The plain form stays at /signup/form.

Login: the "Ask the assistant" chat is on the login page too (e.g. "I forgot my password"). **Changed:** it doesn't read the login form itself; it never sees the password.

4. Student check and the Rs 5 discount

- A clear consent line before the upload: what the photo is for and that it is deleted after the check.
- Accepted: JPG or PNG, up to 5 MB. The photo stays in memory only; it is never written to the database.
- Haiku reads the card: student's name close to the account name, a college name, and a valid-until date in the future. It answers "verified", or asks for a clearer photo ("the date is cut off").
- DeployHub keeps only: **verified student, valid until <date>** (or one year when the card shows no date). After that date the discount stops and the user can verify again.
- Alternative: a code sent to a college email (e.g. name@college.ac.in). Harder to fake than a photo.
- The discount is applied by the backend when it creates the Razorpay order (Rs 44 instead of Rs 49). The browser can't change the amount.

> A photo can be edited, so this is a light check, not proof. The discount is small (Rs 5) to keep the loss from a fake card small, and one account can be verified only once per card validity period.

5. Deploy helper and pop-ups

In the chat, the user can ask anything about DeployHub ("Why does my frontend need a route?"). The assistant uses DeployHub's real data, not guesses: the project's settings, Detect results and "What this app needs".

Pops up when	What it reads	Model	Explains
A deploy fails (build error, crash on start, health check)	Project settings, the failure reason, the last 300 log lines (secrets masked)	Opus	On click; saved, so free next time
"What this app needs" finds required settings missing	The missing names and what they are (never values)	Haiku	On click
A payment fails	Razorpay's error message (no card or bank data)	Haiku	Right away
A database import is refused	The refusal message	Haiku	Right away

The pop-up is a small panel the user can close; it never blocks the page. **Changed:** the expensive Opus explanation and the missing-settings one start on a click, so simply opening a page never spends money. The assistant only explains and suggests. It can't press buttons, change settings or start deploys.

6. How it is built

Flow of one message

Browser chat panel -> Java backend (checks the login and limits, hides secrets) -> Claude API -> reply. **Changed:** the reply comes back in one piece with a typing indicator instead of word by word; Haiku answers in seconds, and it kept the design simpler. The browser keeps the conversation; the server stores no chats.

Backend (Java)

Part	Job
`assistant/GeminiAssistantModel`	Calls Google Gemini (REST, key in a header, never in the URL)
`assistant/ClaudeAssistantModel`	Calls Claude (Anthropic Java SDK 3.2.0) when provider=claude

Part	Job
assistant/AssistantBudget	Daily cap, per-user limits, cost of each call (assistant_usage)
assistant/EmailTypos, EmailProofs	"Did you mean gmail.com?"; the signed 30-minute email proof
assistant/SecretMasker	Replaces env var values, tokens and passwords in logs with ****** before sending
assistant/OfflineReplies	Simple fixed replies when no API key is set or Claude is down
controllers/AssistantController	Chat, stream, student ID upload and explain-failure endpoints
services/.../AssistantService	Sign-up checks and codes, chat, explanations, built-in fallbacks
services/.../EmailCodeService	6-digit codes for sign-up and college emails
services/.../StudentVerificationService	Photo check, college email code, valid-until date
BillingService (changed)	Rs 44 order amount for verified students

Endpoints

Method and path	What it does
GET /api/assistant/status	Public: is the AI available now
POST /api/assistant/signup/check	Public: check a name or email (no AI)
POST /api/assistant/signup/email-code (+ /verify)	Public: send / check the sign-up code
POST /api/assistant/chat	Public: a question (limited per user, or per network before login)
POST /api/assistant/explain	Explain a failed deployment or an error
POST /api/students/verify/id-card	Student ID photo check (multipart, 5 MB, consent)
POST /api/students/verify/college-email/code (+ /college-email)	Send / check a college email code
PATCH /api/auth/me	Change student / working professional

Database (Flyway V15, applied to your Oracle database)

- users: occupation, email_verified_at, student_verified_until, student_verified_by.
- email_codes: hashed 6-digit codes (sign-up and college email).
- assistant_usage: requests, tokens and cost per day and per user or network.
- deployment_explanations: the saved explanation of a failed deployment.
- **Changed:** no chat tables; chats aren't stored at all.

Frontend (Angular + PrimeNG)

- core/components/assistant: the "Ask the assistant" chat (every app page and the login page) and the pop-up tip.
- features/onboarding: the RAY-style sign-up screen at /signup with the "Your details" card.
- Settings > Student discount; the payment page shows the student price.

Settings (application.properties)

Setting	Default
deployhub.assistant.enabled	true
deployhub.assistant.provider	gemini (or claude)
deployhub.assistant.gemini.chat-model / explain-model	gemini-3.5-flash-lite / gemini-3.8-flash
deployhub.assistant.gemini.api-key	${GEMINI_API_KEY:} (from .env)
deployhub.assistant.gemini.free-tier	true (cost 0, no student ID photos)
deployhub.assistant.gemini.id-cards-on-free-tier	false; true only for testing (on in your local .env), use your own or a sample card
deployhub.assistant.claude.chat-model / explain-model	claude-haiku-5-5 / claude-opus-5-5
deployhub.assistant.max-requests-per-user-per-day	50

Setting	Default
`deployhub.assistant.max-requests-per-ip-per-day`	20 (before signing in)
`deployhub.assistant.daily-budget-inr`	20 (everyone together; then built-in replies until midnight)
`deployhub.billing.student-discount`	500 (paise = Rs 5)

7. Privacy: what the AI sees and what is kept

Data	Sent to the AI?	Kept by DeployHub?
Name, email	Yes, to check and suggest	Yes (the account)
Password	**Never**	Only the BCrypt hash, as today
Student ID photo	Paid tiers only, once, for the check (never on Gemini's free tier)	**No**: deleted after the check; only the valid-until date
Deploy logs	Yes, with secrets masked	Logs as today; chats are not stored
Env var values, access tokens, DB passwords	**Never** (masked)	Encrypted, as today
Payment card / bank data	**Never** (DeployHub never has it)	No (Razorpay holds it)

Why no Aadhaar

- The Aadhaar Act and the Supreme Court's 2018 judgment limit private companies; it can't be made compulsory.
- UIDAI rules require stored Aadhaar numbers to be kept in a special Aadhaar Data Vault.
- The DPDP Act 2023 allows fines up to Rs 250 crore for personal data leaks.
- A card photo (number, address, date of birth, face) is very valuable to fraudsters if leaked.
- Razorpay already checks the identity of people who pay. (General information, not legal advice.)

8. Cost

Model	Price per million tokens (in / out)	One sign-up chat	Explaining a failed deploy
Gemini free tier (default)	free, within Google's daily limits	Rs 0	Rs 0
Gemini paid (3.5 Flash-Lite / 3.8 Flash)	$0.30 / $2.50 and $0.75 / $3.75 (from 2027: $1.50 / $7.50)	about Rs 0.5	about Rs 1-2
Haiku 5.5	$0.10 / $0.50	about Rs 0.15	about Rs 0.25
Opus 5.5	$4 / $20	about Rs 5	about Rs 10

Each conversation uses **one** model; costs don't add up. Example: 100 sign-ups and 20 failed deploys.

Choice	100 sign-ups	20 failed deploys	Total
Both (chosen): Haiku chat, Opus failures	Rs 15	Rs 200	**about Rs 215**
Only Opus	Rs 500	Rs 200	about Rs 700
Only Haiku	Rs 15	Rs 5	about Rs 20

Estimates only; real cost depends on chat length and log size. The provider and models can be switched with one setting, no code change. On Gemini's free tier, when Google's free daily limit is reached, Gemini answers "too many requests" and DeployHub gives the built-in answers until the limit resets.

> **Daily cap: Rs 20** (decided 28 October 2026, while DeployHub has no users yet; on Gemini's free tier nothing is spent, so only the per-user limits apply). On Claude that is roughly 130 Haiku sign-up chats, or about 2 Opus failed-deploy explanations plus some chats, per day. When the day's Rs 20 is used up, the assistant switches to the offline replies until midnight (India time); DeployHub itself keeps working. Raise it with one setting when real users arrive.

9. Setting up the API key (your steps)

Gemini (free, the default)

- Open **aistudio.google.com**, sign in with a Google account, click **Get API key** and create a key.

- Add a line `GEMINI_API_KEY=...` to deployhub-backend/.env yourself. Don't paste the key in chat or commit it to git.
- Restart the backend. Without a key, the assistant uses the simple built-in replies.
- Don't add billing unless you want to: the free tier is enough for testing. If you do add billing, set `deployhub.assistant.gemini.free-tier=false` (turns the student photo check on and counts the cost against the daily cap).

Claude (paid, optional)

- console.anthropic.com: add prepaid credit, create a key, put ANTHROPIC_API_KEY=... in .env, set `deployhub.assistant.provider=claude`, restart. Set a monthly spend limit there.

10. Risks and fixes

Problems we expect for an AI assistant like this, and how the design handles each one. Severity is how bad it would be without the fix.

ID	Severity	Risk	Fix in the design
AI-01	High	Prompt injection: a repo file, log line or chat message says "ignore your rules and ..."	Logs and repo text are passed as data, clearly marked, never as instructions. The AI has no tools that change anything: it can only reply with text, so the worst case is a wrong sentence, not a wrong action.
AI-02	High	Secrets leak to the AI through deploy logs (a password printed by the app)	SecretMasker replaces every env var value, access token and DB password of the project, plus common key patterns (sk-..., rzp_..., AKIA..., Bearer ...), with ****** before anything is sent. Tested with real-looking logs.
AI-03	High	Cost runs away (a bot chats all day, or a huge log is sent)	Per-user and per-IP message limits, a daily budget cap, a maximum log size (last 300 lines), max_tokens on every reply, a spend limit in the provider's console; on Gemini's free tier nothing is charged.
AI-04	Medium	Fake student ID to get the discount	Light AI check + college email alternative; discount only Rs 5; one verification per account per validity period; amount decided by the backend, not the browser.
AI-05	Medium	The AI gives wrong advice ("hallucination")	It is given DeployHub's real data (Detect, What this app needs, the build hint) and told to say "I'm not sure" when unsure. Replies are labelled as suggestions; the user makes every change.
AI-06	Medium	No API key, the AI down or busy (503), or rate-limited (429)	A busy model is retried twice, then the lighter chat model answers instead (BUG-35); after that, built-in replies give fixed, useful messages, with an "Ask the AI again" button. The normal forms keep working.
AI-07	Medium	Sign-up chat abused before login (spam accounts, emails flooded)	Same IP limits as registration and Forgot password (audit_events), email codes limited per email and IP.
AI-08	Medium	Student ID photo kept by mistake (privacy)	Photo is held in memory only, sent once, then dropped; never written to disk or the database. A test checks nothing is stored.
AI-09	Low	Photo problems: too big, not an image, location data inside (EXIF)	5 MB limit, JPG/PNG check by file content (not just the name), metadata removed before sending.
AI-10	Low	The AI refuses to answer (stop reason "refusal") or the reply is cut off	Checked on every reply; the user sees a friendly fallback message instead of a blank chat. Fallback models enabled on the API.
AI-11	Low	Opus replies feel slow on failed deploys	A "Reading what happened..." indicator; the explanation is saved, so it is slow only once.
AI-12	Low	Emails with codes aren't delivered	Known today: without MAIL_USERNAME/MAIL_PASSWORD in .env, codes go to the backend log. Gmail sending must be set up before real users sign up.
AI-13	Low	Pop-ups annoy users	Small, closable panel next to the problem; the expensive ones explain only on a click.

> **Existing open bug:** BUG-21 (sign-up on the demo copy with H2) is still open and is not related to the assistant. The AI assistant needs internet access to the Claude API; the demo copy may not have it, so it would use the offline replies there.

11. Bugs found while building (all fixed, with tests)

ID	Bug	Fix
BUG-32	/api/auth/me answered 503 during a database outage (it now reads the profile from the database), while it used to keep working from the session	Falls back to the session's name and email while the database is down
BUG-33	SMTP_PASS=... (and other short secret names) was not masked before going to the AI	Extra masking rule for names with pass, pwd, secret, token, key, credential, auth
BUG-34	A correct but uncommon email domain (mail.com) was "corrected" to gmail.com and blocked sign-up	Real domains are never corrected, and a suggestion is only a suggestion: the user can keep their address

12. How it was tested

- Backend: 375 tests pass (63 new) with a fake AI, so no money was spent; the Gemini client is also tested against a fake Google server (what it sends, the key in a header, refusals, used-up quota, the ID card JSON): sign-up checks and codes, proofs, chat, limits, the Rs 20 cap, outages, refusals, masking, saved explanations, ownership, student photo checks (name, date, consent, 5 per day), metadata removal, college email, the Rs 44 order amount.
- Frontend: 157 tests pass (16 new): the sign-up conversation, typo buttons, questions, the chat, the pop-up, Settings > Student discount.
- On your real Oracle database: V15 applied, then a full sign-up with an email code through the running app (emailVerified true) and a chat with built-in answers; the test accounts were deleted afterwards.
- **Not tested yet:** real Gemini or Claude answers, because there is no key yet. With a free Gemini key, trying a few real questions costs nothing.

13. Still open

- Your free Gemini API key in .env (needed only for real AI replies).
- Gmail sending (MAIL_USERNAME / MAIL_PASSWORD) before real users sign up.
- A few real questions once the key is in (free on Gemini), to check the answers' quality.